
ST3101: 4. Exercises on random vectors

1. Expected value of vector

Define the random vector

$$\mathbf{x} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

and assume that $E(X_1) = 1$, $E(X_2) = 3$, and $E(X_3) = 0$.

(a) What is $E(\mathbf{x})$?

(b) Let $Y_1 = 2X_1$, $Y_2 = -X_2$, and $Y_3 = 3X_3$, and define

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \end{bmatrix}$$

What is $E(\mathbf{y})$?

(c) Find the matrix $\mathbf{A}$ such that

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

(a) By the definition of expectations for vectors

$$E(\mathbf{x}) = \begin{bmatrix} E(X_1) \\ E(X_2) \\ E(X_3) \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$$

(b) By the definition of expectations for vectors

$$E(\mathbf{y}) = \begin{bmatrix} E(Y_1) \\ E(Y_2) \\ E(Y_3) \end{bmatrix} = \begin{bmatrix} E(2X_1) \\ E((-1)X_2) \\ E(3X_3) \end{bmatrix} = \begin{bmatrix} 2E(X_1) \\ -E(X_2) \\ 3E(X_3) \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 0 \end{bmatrix}$$

(c) If we think about what $\mathbf{y} = \mathbf{A}\mathbf{x}$ is,

$$\mathbf{y} = \underbrace{\begin{bmatrix} | & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \\ | & | & | \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}}_{\mathbf{x}} = X_1 \times \begin{bmatrix} | \\ \mathbf{a}_1 \\ | \end{bmatrix} + X_2 \times \begin{bmatrix} | \\ \mathbf{a}_2 \\ | \end{bmatrix} + X_3 \times \begin{bmatrix} | \\ \mathbf{a}_3 \\ | \end{bmatrix}$$

and since $Y_i = a_{ii}X_i$, for $i = 1, 2, 3$, column 1 must have the first element $a_{11} = 2$, and the rest zero; column 2 must have the second element $a_{22} = -1$, and the rest zero; and, column 3 must have the third element $a_{33} = 3$, and the rest zero. Thus

$$\mathbf{y} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}}_{\mathbf{x}} = X_1 \times \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + X_2 \times \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} + X_3 \times \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 2X_1 \\ -X_2 \\ 3X_3 \end{bmatrix}$$

and $\mathbf{A}$ is a diagonal matrix.

2. Expected value of linear function of vector

The random variable $\mathbf{x}$ has expected value

$$E(\mathbf{x}) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(a) Let $\mathbf{A}$ be a diagonal 4×4 matrix, with diagonal elements 1, 2, 3, and 4. Defining

$$\mathbf{y} = \mathbf{A}\mathbf{x},$$

what is $E(\mathbf{y})$?

(b) Let $\mathbf{A}$ be an arbitrary 4×4 matrix, what is $E(\mathbf{A}\mathbf{x})$?

(c) For $\mathbf{A}$ as in 2(a), and

$$\mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ 2 \\ 0 \end{bmatrix}$$

define

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b},$$

what is $E(\mathbf{y})$?

(a) Since $Y_i = a_{ii}X_i$, for all i , and

$$E(Y_i) = E(a_{ii}X_i) = a_{ii}E(X_i) = 0$$

$E(\mathbf{y})$ is a vector of zeros

(b) We could write this out in terms of the elements, which by matrix multiplication will be

$$E(Y_i) = E\left(\sum_{j=1}^4 a_{ij}X_j\right) = \sum_{j=1}^4 a_{ij}E(X_j) = 0$$

or we may use the result that for $\mathbf{y} = \mathbf{A}\mathbf{x}$

$$E(\mathbf{y}) = E(\mathbf{A}\mathbf{x}) = \mathbf{A}E(\mathbf{x}) = \mathbf{0}$$

where the last equality follows from the fact that anything multiplied by a vector of zeros is a vector of zeros.

(c) Again, we could write this out in terms of the elements, which by matrix multiplication will be

$$E(Y_i) = E\left(\sum_{j=1}^4 a_{ij}X_j + b_i\right) = \sum_{j=1}^4 a_{ij}E(X_j) + b_i = 0 + b_i$$

and thus

$$E(\mathbf{y}) = \mathbf{0} + \mathbf{b} = \begin{bmatrix} 1 \\ 5 \\ 2 \\ 0 \end{bmatrix}$$

or we may use the result that for $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b}$

$$E(\mathbf{y}) = E(\mathbf{A}\mathbf{x} + \mathbf{b}) = E(\mathbf{A}\mathbf{x}) + E(\mathbf{b}) = \mathbf{A}E(\mathbf{x}) + \mathbf{b} = \mathbf{0} + \mathbf{b} = \mathbf{b}$$

3. More on expectations of linear functions

Let $\mathbf{x}$ be a 3×1 vector of random variables with For $\mathbf{A}$ as in 2(a), and denote $E(\mathbf{x}) = \boldsymbol{\mu}$. If $\mathbf{y} = \mathbf{A}\mathbf{x}$ and

$$\mathbf{A} = \begin{bmatrix} | & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{0} \\ | & | & | \end{bmatrix}$$

i.e., where the last column is all zeros.

(a) What is the difference in $E(Y_3)$ when μ_3 is equal to 1 and -1 , respectively?

(b) In terms of μ_i and elements a_{ij} of $\mathbf{A}$, what is $E(\mathbf{y})$?

(c) For $\mathbf{y} = \mathbf{A}^\top \mathbf{x}$, what is $E(Y_3)$?

(a) Since

$$E(\mathbf{y}) = E \left(\begin{bmatrix} \begin{bmatrix} | \\ | \\ | \end{bmatrix} & \begin{bmatrix} | \\ | \\ | \end{bmatrix} & \begin{bmatrix} | \\ | \\ | \end{bmatrix} \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \right) = E(X_1) \begin{bmatrix} | \\ | \\ | \end{bmatrix} + E(X_2) \begin{bmatrix} | \\ | \\ | \end{bmatrix} + E(X_3) \begin{bmatrix} | \\ | \\ | \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

the expected value $E(\mathbf{y})$ does not depend on $\mu_3 = E(X_3)$.

(b)

$$E(Y_i) = a_{i1}\mu_1 + a_{i2}\mu_2$$

(c) 0

4. Expectations of sums of vectors

Let $\mathbf{y}$ and $\mathbf{x}$ be two random vectors with

$$E(\mathbf{x}) = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \text{ and } E(\mathbf{y}) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Let the random variable

$$\mathbf{z} = \mathbf{x} + \mathbf{y}$$

(a) What is $E(\mathbf{z})$?

(b) For

$$\mathbf{a} = \begin{bmatrix} 0.5 \\ 1 \\ -0.5 \end{bmatrix}$$

what is $E(\mathbf{a}^\top \mathbf{z})$?

(c) What is $E(\mathbf{z} \mathbf{a}^\top)$?

(d) With the same $\mathbf{a}$ but $\mathbf{z} = \mathbf{x} - \mathbf{y}$, what is $E(\mathbf{a}^\top \mathbf{z})$?

(a)

$$E(\mathbf{z}) = E(\mathbf{x} + \mathbf{y})$$

We know now that this is a matrix with expected values, where for each element, we have

$$E(Z_i) = E(X_i + Y_i) = E(X_i) + E(Y_i)$$

but this is exactly what we get if we add the vectors $E(\mathbf{x})$ and $E(\mathbf{y})$

$$E(\mathbf{z}) = E(\mathbf{x}) + E(\mathbf{y}) = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix}$$

(b) We know that

$$E(\mathbf{a}^\top \mathbf{z}) = \mathbf{a}^\top E(\mathbf{z}) = \mathbf{a}^\top (E(\mathbf{x}) + E(\mathbf{y}))$$

but we can do it in terms of elements to convince ourselves that we are allowed to do this.

$$\mathbf{a}^\top \mathbf{z} = \sum_{i=1}^3 a_i Z_i = \sum_{i=1}^3 a_i (X_i + Y_i) = \sum_{i=1}^3 a_i X_i + \sum_{i=1}^3 a_i Y_i$$

and thus

$$\begin{aligned} E(\mathbf{a}^\top \mathbf{z}) &= E\left(\sum_{i=1}^3 a_i X_i + \sum_{i=1}^3 a_i Y_i\right) \\ &= \sum_{i=1}^3 a_i E(X_i) + \sum_{i=1}^3 a_i E(Y_i) \\ &= \sum_{i=1}^3 a_i (E(X_i) + E(Y_i)) \\ &= \mathbf{a}^\top (E(\mathbf{x}) + E(\mathbf{y})) \end{aligned}$$

We calculated $E(\mathbf{z})$ before

$$E(\mathbf{a}^\top \mathbf{z}) = \begin{bmatrix} 0.5 & 1 & -0.5 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix} = (.5)(5) + (1)(6) + (-.5)(7) = 5$$

(c)

$$\begin{aligned} E(\mathbf{z} \mathbf{a}^\top) &= E(\mathbf{z}) \mathbf{a}^\top \\ &= \begin{bmatrix} 5 \\ 6 \\ 7 \end{bmatrix} \begin{bmatrix} 0.5 & 1 & -0.5 \end{bmatrix} \\ &= \begin{bmatrix} 5 \times .5 & 5 \times 1 & 5 \times (-.5) \\ 6 \times .5 & 6 \times 1 & 6 \times (-.5) \\ 7 \times .5 & 7 \times 1 & 7 \times (-.5) \end{bmatrix} \\ &= \begin{bmatrix} 2.5 & 5 & -2.5 \\ 3 & 6 & -3 \\ 3.5 & 7 & -3.5 \end{bmatrix} \end{aligned}$$

(d) We know that

$$E(\mathbf{a}^\top \mathbf{z}) = \mathbf{a}^\top E(\mathbf{z}) = \mathbf{a}^\top (E(\mathbf{x}) - E(\mathbf{y}))$$

but we can do it in terms of elements to convince ourselves that we are allowed to do this.

$$E(\mathbf{a}^\top \mathbf{z}) = E\left(\sum_{i=1}^3 a_i Z_i\right) = \sum_{i=1}^3 a_i (X_i - Y_i) = \sum_{i=1}^3 a_i E(X_i) - \sum_{i=1}^3 a_i E(Y_i)$$

$$\begin{aligned} E(\mathbf{a}^\top \mathbf{z}) &= \begin{bmatrix} 0.5 & 1 & -0.5 \end{bmatrix} \left(\begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0.5 & 1 & -0.5 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \\ &= (.5)(3) + (1)(4) + (-.5)(5) \\ &= 3 \end{aligned}$$

5. Covariance

Consider the variable $\mathbf{x}$ with expected value

$$E(\mathbf{x}) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

with variance-covariance matrix

$$V(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

(a) Using the definition

$$V(\mathbf{x}) = E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^\top]$$

show that

$$V(\mathbf{x}) = E(\mathbf{x}\mathbf{x}^\top)$$

(b) Let

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

for

$$\mathbf{A} = \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix}$$

What is $V(\mathbf{y})$?

(c) Using the definition

$$Cov(\mathbf{x}, \mathbf{y}) = E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{y} - E(\mathbf{y}))^\top]$$

find $Cov(\mathbf{x}, \mathbf{y})$.

(a)

$$\begin{aligned} V(\mathbf{x}) &= E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^\top] \\ &= E\left[\left(\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right)\left(\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}\right)^\top\right] \\ &= E\left[\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}\begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}^\top\right] \\ &= E(\mathbf{x}\mathbf{x}^\top) \end{aligned}$$

We could also use the rearrangement

$$\begin{aligned} V(\mathbf{x}) &= Cov(\mathbf{x}, \mathbf{x}) \\ &= E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^\top] \\ &= E[(\mathbf{x} - E(\mathbf{x}))\mathbf{x}^\top - (\mathbf{x} - E(\mathbf{x}))E(\mathbf{x})^\top] \\ &= E[\mathbf{x}\mathbf{x}^\top - E(\mathbf{x})\mathbf{x}^\top - \mathbf{x}E(\mathbf{y})^\top + E(\mathbf{x})E(\mathbf{x})^\top] \\ &= E[\mathbf{x}\mathbf{x}^\top - E(\mathbf{x})\mathbf{x}^\top - \mathbf{x}E(\mathbf{x})^\top + E(\mathbf{x})E(\mathbf{x})^\top] \\ &= E[\mathbf{x}\mathbf{x}^\top] - E(\mathbf{x}) \underbrace{E[\mathbf{y}^\top]}_{E(\mathbf{x}^\top)=E(\mathbf{x})^\top} - E[\mathbf{x}]E(\mathbf{x})^\top + E(\mathbf{x})E(\mathbf{x})^\top \\ &= E(\mathbf{x}\mathbf{x}^\top) - E(\mathbf{x})E(\mathbf{x})^\top \underset{=0}{} \end{aligned}$$

(b) Using formula for the covariance of a linear combination

$$V(\mathbf{y}) = V(\mathbf{Ax}) = \mathbf{A}V(\mathbf{x})\mathbf{A}^\top$$

Let us start with $\mathbf{A}V(\mathbf{x})$

$$\begin{aligned}\mathbf{A}V(\mathbf{x}) &= \begin{bmatrix} 2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \\ &= \left[\begin{bmatrix} 2 \times 2 \\ 5 \times 1 \end{bmatrix} + 1 \times \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \times 2 \\ 2 \times 1 \end{bmatrix} + 2 \times \begin{bmatrix} 3 \\ 1 \end{bmatrix} \right] \\ &= \begin{bmatrix} 4+3 & 2+6 \\ 10+1 & 5+2 \end{bmatrix} \\ &= \begin{bmatrix} 7 & 8 \\ 11 & 7 \end{bmatrix}\end{aligned}$$

Now do the last postmultiplication by $\mathbf{A}^\top$

$$\begin{aligned}\mathbf{A}V(\mathbf{x})\mathbf{A}^\top &= \begin{bmatrix} 7 & 8 \\ 11 & 7 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix} \\ &= \left[\begin{bmatrix} 2 \times 7 \\ 3 \times 11 \end{bmatrix} + 5 \times \begin{bmatrix} 8 \\ 7 \end{bmatrix} \quad \begin{bmatrix} 5 \times 7 \\ 1 \times 11 \end{bmatrix} + 1 \times \begin{bmatrix} 8 \\ 7 \end{bmatrix} \right] \\ &= \begin{bmatrix} 14+24 & 35+8 \\ 22+21 & 55+7 \end{bmatrix} \\ &= \begin{bmatrix} 38 & 43 \\ 43 & 62 \end{bmatrix}\end{aligned}$$

We see that this is an admissible covariance matrix - the variances (38 and 62) are positive and the matrix is symmetric.

(c) First note that

$$E(\mathbf{y}) = E(\mathbf{A}\mathbf{x}) = \mathbf{A}E(\mathbf{x}) = \mathbf{0}$$

so that we have that

$$\text{Cov}(\mathbf{x}, \mathbf{y}) = E[(\mathbf{x} - E(\mathbf{x})(\mathbf{y} - E(\mathbf{y})))^\top] = E(\mathbf{x}\mathbf{y}^\top)$$

Writing out $\mathbf{y} = \mathbf{A}\mathbf{x}$

$$E(\mathbf{x}\mathbf{y}^\top) = E(\mathbf{x}(\mathbf{A}\mathbf{x})^\top) = E(\mathbf{x}\mathbf{x}^\top \mathbf{A}^\top) = E(\mathbf{x}\mathbf{x}^\top) \mathbf{A}^\top = V(\mathbf{x}) \mathbf{A}^\top$$

The last step is justified as $\mathbf{A}$ is a matrix with elements that are constant. Do we have to calculate this anew? No, note that we previously have calculated $\mathbf{A}V(\mathbf{x})$ and

$$(\mathbf{A}V(\mathbf{x}))^\top = V(\mathbf{x})^\top \mathbf{A}^\top$$

Now, since $V(\mathbf{x})$ is symmetric, $V(\mathbf{x})^\top = V(\mathbf{x})$, and

$$V(\mathbf{x})^\top \mathbf{A}^\top = V(\mathbf{x}) \mathbf{A}^\top$$

and our answer is

$$V(\mathbf{x}) \mathbf{A}^\top = \begin{bmatrix} 7 & 8 \\ 11 & 7 \end{bmatrix}^\top = \begin{bmatrix} 7 & 11 \\ 8 & 7 \end{bmatrix}$$

6. Normal vectors

Consider the variables $X \sim N(0, 1)$, $Y \sim N(-1, 3)$, and $Z \sim N(0.5, 0.5)$, and define

$$\mathbf{x} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Let

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 1 & 3 & 6 \end{bmatrix}$$

(a) Consider

$$\mathbf{u} = \mathbf{A}\mathbf{x}$$

and find $E(\mathbf{u})$

(b) Assume that X , Y , and Z are *linearly independent* - what is $V(\mathbf{u})$ and what distribution does the first element U_1 follow?

(a)

$$\begin{aligned} E(\mathbf{u}) &= E\left(\begin{bmatrix} 2X + Y + 2Z \\ X + 3Y + 6Z \end{bmatrix}\right) \\ &= \begin{bmatrix} E(2X) + E(Y) + E(2Z) \\ E(X) + E(3Y) + E(6Z) \end{bmatrix} \\ &= \begin{bmatrix} 2E(X) + E(Y) + 2E(Z) \\ E(X) + 3E(Y) + 6E(Z) \end{bmatrix} \\ &= \begin{bmatrix} -1 + 1 \\ -3 + 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

We could check that our standard formula for expectations work, namely

$$E(\mathbf{u}) = E(\mathbf{A}\mathbf{x}) = \mathbf{A}E(\mathbf{x}) = 0 \times \begin{bmatrix} 2 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ 3 \end{bmatrix} + .5 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(b)

$$V(\mathbf{u}) = V(\mathbf{Ax}) = \mathbf{AV}(\mathbf{x})\mathbf{A}^\top$$

In $\mathbf{AV}(\mathbf{x})$, is easy because

$$V(\mathbf{x}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}$$

is diagonal so that the first column of $\mathbf{AV}(\mathbf{x})$ will be

$$1 \times \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 0 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 0 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

the second column

$$3 \times \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

and the third

$$.5 \times \begin{bmatrix} 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Giving us

$$\mathbf{AV}(\mathbf{x}) = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 9 & 3 \end{bmatrix}$$

Further

$$\mathbf{AV}(\mathbf{x})\mathbf{A}^\top = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 9 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 2 \times 2 + 3 + 2 \times 1 & 2 + 3 \times 3 + 6 \\ 2 + 9 + 2 \times 3 & 1 + 3 \times 9 + 6 \times 3 \end{bmatrix}$$

Thus

$$V(\mathbf{u}) = \begin{bmatrix} 9 & 17 \\ 17 & 46 \end{bmatrix}$$

Looking closely at U_1

$$U_1 = \begin{bmatrix} 2 & 1 & 2 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = 2X + Y + 2Z$$

We have earlier calculated $E(\mathbf{u}) = \mathbf{0}$ and above

$$V(U_1) = V(2X + Y + 2Z) = 2^2V(X) + V(Y) + 2^2V(Z) = 4 + 3 + 4/2 = 9$$

but we note that U_1 is the weighted sum of 3 normal variates and thus $U_1 \sim N(0, 9)$.

7. Linear functions of Normal vectors

Let

$$\mathbf{x} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

where

$$\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

and $\boldsymbol{\Sigma} = \mathbf{I}$. Further, define the matrices

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

- (a) What is $E(\mathbf{Ax} + \mathbf{b})$?
 (b) What is $V(\mathbf{Ax} + \mathbf{b})$?
 (c) Based on properties of linear combinations of normal variates, show that the first element of $\mathbf{Ax} + \mathbf{b}$ follows a normal distribution.

(a)

$$E(\mathbf{Ax} + \mathbf{b}) = \underbrace{E(\mathbf{Ax})}_{\text{exp of const.}\mathbf{x}} + \underbrace{E(\mathbf{b})}_{\text{const.}} = \mathbf{A} \underbrace{E(\mathbf{x})}_{\boldsymbol{\mu}=\mathbf{0}} + \mathbf{b} = \mathbf{b}$$

(b) First, use rule for variances of linear combinations

$$V(\mathbf{Ax} + \underbrace{\mathbf{b}}_{\text{cons.}}) = V(\mathbf{Ax}) = \mathbf{A} \underbrace{V(\mathbf{x})}_{=\mathbf{I}} \mathbf{A}^\top = \mathbf{AA}^\top$$

The variance of $\mathbf{Ax} + \mathbf{b}$ is clearly $\mathbf{A}$ post-multiplied by its transpose, but $\mathbf{A}$ is also diagonal. For diagonal matrices $\mathbf{D}$, we know that $\mathbf{BD}$, means multiplying the columns of $\mathbf{B}$ with the diagonal elements of $\mathbf{D}$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Even more straightforward is to recognise that $\mathbf{A} = 2\mathbf{I}$, and thus

$$V(\mathbf{Ax} + \mathbf{b}) = V(2\mathbf{x}) = 4\mathbf{I}$$

We can double check this by noting that $\mathbf{y} = \mathbf{Ax} + \mathbf{b}$, will have elements $Y_i = 2X_i + 1$, giving $E(Y_i) = 1$ and $V(Y_i) = V(2X_i + 1) = 2^2 V(X_i) = 4$.

(c) From (b), we have that $Y_i = 2X_i + 1$, and all X_i are iid $N(0, 1)$. Consequently, $Y_1 \sim N(1, 2^2)$. The vector $\mathbf{y}$ will have iid normal variates in the rows, making it a multivariate normal distribution

$$\mathbf{y} \sim N(\mathbf{1}, 4\mathbf{I})$$

8. More linear functions of Normal vectors

Let

$$\mathbf{x} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

where

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ -0.5 \\ 3 \end{bmatrix}$$

and

$$\boldsymbol{\Sigma} = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Further, define the matrix

$$\mathbf{A} = \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(a) What is the correlation $\text{corr}(X_1, X_3)$?

(b) What is the correlation $\text{corr}(X_1, X_2)$?

(c) If

$$\mathbf{y} = \mathbf{A}\mathbf{x}$$

what is $V(\mathbf{y})$?

(d) If

$$\mathbf{y} = \mathbf{x} - \boldsymbol{\mu}$$

what is $E(\mathbf{y})$?

(e) If

$$\mathbf{y} = \mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}$$

what is $V(\mathbf{y})$ and $E(\mathbf{y})$?

(f) If

$$\mathbf{y} = \mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}$$

what is the correlation $\text{corr}(Y_1, Y_3)$?

(a) We use the definition

$$\text{corr}(X_i, X_j) = \frac{\text{Cov}(X_i, X_j)}{\sqrt{V(X_i)V(X_j)}}$$

For $i = 1$ and $j = 3$, we have $\text{Cov}(X_1, X_3) = 0$ and hence the correlation is zero.

(b)

$$\text{corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{V(X_1)V(X_2)}} = \frac{4}{\sqrt{5 \times 5}} = \frac{4}{5}$$

(c) Use the now familiar rule that $V(\mathbf{A}\mathbf{x}) = \mathbf{A}V(\mathbf{x})\mathbf{A}^\top$. We start with $\mathbf{A}V(\mathbf{x})$, doing it column by column

$$\mathbf{A} \begin{bmatrix} | \\ | \\ | \end{bmatrix} = \sigma_1^2 \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ 0 \end{bmatrix} + \sigma_{12} \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} + \sigma_{13} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5(\frac{2}{3}) + 4(-\frac{1}{3}) \\ 5(-\frac{1}{3}) + 4(\frac{2}{3}) \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\mathbf{A} \begin{bmatrix} | \\ | \\ | \end{bmatrix} = \sigma_{12} \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ 0 \end{bmatrix} + \sigma_2^2 \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} + \sigma_{23} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4\frac{2}{3} + 5(-\frac{1}{3}) \\ 4(-\frac{1}{3}) + 5(\frac{2}{3}) \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$\mathbf{A} \begin{bmatrix} | \\ | \\ | \end{bmatrix} = \sigma_{13} \begin{bmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ 0 \end{bmatrix} + \sigma_{23} \begin{bmatrix} -\frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix} + \sigma_3^2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Now that we have $\mathbf{A}V(\mathbf{x})$, we need $\mathbf{A}V(\mathbf{x})\mathbf{A}^\top$

$$\mathbf{A}V(\mathbf{x})\mathbf{A}^\top = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{2}{3} & -\frac{1}{3} & 0 \\ -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

the first column of which will be (note that we do not need the last column as the last row of the first column of $\mathbf{A}$ is 0)

$$\begin{pmatrix} \frac{2}{3} \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{1}{3} \\ 2 \\ 0 \end{pmatrix} = \begin{bmatrix} \frac{4}{3} - \frac{1}{3} \\ \frac{2}{3} + \frac{2}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

For the second column

$$\begin{pmatrix} -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{2}{3} \\ 2 \\ 0 \end{pmatrix} = \begin{bmatrix} -\frac{2}{3} + \frac{2}{3} \\ -\frac{1}{3} + \frac{4}{3} \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

and, finally, for the third column we will simply get

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The variance of $\mathbf{y} = \mathbf{A}\mathbf{x}$ is clearly

$$V(\mathbf{y}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{I}$$

(d)

$$E(\mathbf{y}) = E(\mathbf{x} - \boldsymbol{\mu}) = E(\mathbf{x}) - \boldsymbol{\mu} = \boldsymbol{\mu} - \boldsymbol{\mu} = \mathbf{0}$$

(e)

$$E(\mathbf{y}) = E(\mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}) = \mathbf{A}E(\mathbf{x}) - \mathbf{A}\boldsymbol{\mu} = \mathbf{A}\boldsymbol{\mu} - \mathbf{A}\boldsymbol{\mu} = \mathbf{0}$$

and

$$V(\mathbf{y}) = V(\mathbf{A}\mathbf{x} - \mathbf{A}\boldsymbol{\mu}) = V(\mathbf{A}\mathbf{x}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \mathbf{I}$$

(f) From (e) we have that $V(\mathbf{y}) = \mathbf{I}$, hence $Cov(Y_1, Y_3) = 0$ and $corr(Y_1, Y_3) = 0$. In fact Y_1 , Y_2 , and Y_3 are iid standard normal variates $N(0, 1)$